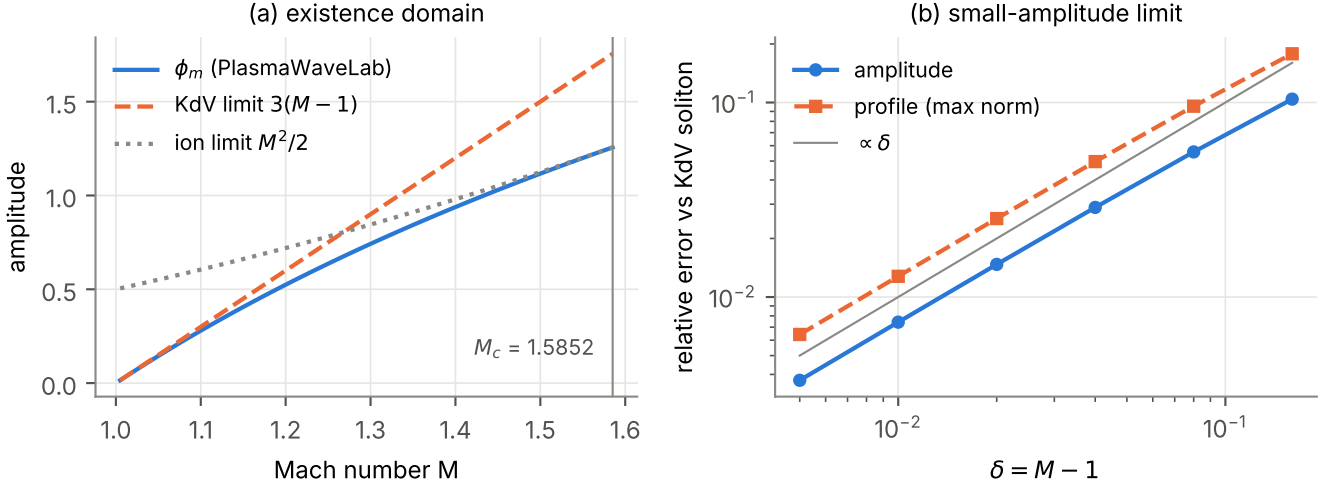
(a) existence domain
ϕ_m (PlasmaWaveLab)
KdV limit $3(M-1)$
ion limit $M^2/2$
M_c = 1.5852
amplitude
Mach number M
0.0
0.5
1.0
1.5
1.0
1.1
1.2
1.3
1.4
1.5
1.6
(b) small-amplitude limit
amplitude
profile (max norm)
$\propto \delta$
relative error vs KdV soliton
$\delta = M - 1$
10^{-1}
10^{-2}